

CARMAN, MB, Canada

WMO: 711470

Lat: 49.500N

Lon: 98.030W

Elev: 268

StdP: 98.14

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.7	(c) -28.0	(d) -34.2	(e) 0.2	(f) -30.6	(g) -31.4	(h) 0.2	(i) -27.8	(j) 11.1	(k) -10.5	(l) 10.2	(m) -11.0	(n) 2.6	(o) 300	(p) 0.666

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	30.3	21.5	28.6	20.7	27.1	19.5	24.0	28.0	22.4	26.5	21.1	25.1	5.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.1	26.5	21.1	16.2	25.1	19.7	14.9	23.5	73.6	27.9	67.4	26.5	62.5	25.3	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.4	9.2	8.3	-35.1	34.0	3.2	1.6	-37.4	35.1	-39.2	36.0	-41.0	36.9	-43.3	38.1
			-34.8	26.3	3.1	1.7	-37.1	27.5	-38.9	28.5	-40.6	29.5	-42.8	30.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-15.1	-13.0	-5.8	4.1	11.1	16.9	19.5	18.5	13.5	5.5	-3.9	-12.1	(6)
(7)		DBStd	13.67	8.21	7.58	7.57	6.01	4.81	3.54	2.92	3.19	4.50	4.87	6.69	7.71	(7)
(8)		HDD10.0	3423	779	643	490	190	46	1	0	0	17	155	417	684	(8)
(9)		HDD18.3	5604	1037	877	747	427	229	67	21	37	154	397	667	943	(9)
(10)		CDD10.0	998	0	0	1	13	79	207	296	264	122	17	0	0	(10)
(11)		CDD18.3	137	0	0	0	0	4	23	58	43	9	0	0	0	(11)
(12)		CDH23.3	1633	0	0	0	5	98	272	579	524	146	9	0	0	(12)
(13)		CDH26.7	426	0	0	0	0	25	61	141	161	36	2	0	0	(13)
(14)	Wind	WSAvg	4.0	4.2	4.1	4.2	4.5	4.5	3.9	3.3	3.4	3.8	4.2	4.1	4.1	(14)
(15)	Precipitation	PrecAvg	522	18	16	25	31	77	91	70	60	45	37	26	25	(15)
(16)		PrecMax	672	38	60	67	76	148	180	160	135	94	158	79	53	(16)
(17)		PrecMin	364	5	0	4	6	25	32	15	14	3	5	3	5	(17)
(18)		PrecStd	88	9	12	17	17	34	38	35	34	21	32	21	13	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	5.4	14.9	23.8	30.0	30.8	31.9	32.9	30.3	24.3	14.7	5.6	(19)
(20)			MCWB	3.0	3.1	9.2	12.9	16.4	22.2	24.7	22.2	19.3	15.5	9.7	2.9	(20)
(21)		2%	DB	2.5	2.7	9.7	20.1	25.9	28.4	29.8	30.5	27.2	19.8	10.6	2.7	(21)
(22)			MCWB	1.0	1.3	6.5	11.3	14.5	19.4	23.1	21.2	18.1	12.8	6.6	1.2	(22)
(23)		5%	DB	-0.2	0.7	6.1	17.3	23.3	26.4	28.2	28.3	24.7	16.6	7.5	0.2	(23)
(24)			MCWB	-1.0	0.0	4.0	10.1	13.9	18.6	21.7	20.5	17.2	11.4	4.9	-0.7	(24)
(25)		10%	DB	-3.1	-1.4	3.3	14.2	20.5	24.6	26.6	26.4	21.8	13.6	4.7	-1.6	(25)
(26)			MCWB	-3.8	-2.2	1.9	8.8	12.9	18.0	20.7	19.4	15.6	9.7	3.1	-2.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	3.5	10.7	15.0	19.2	25.0	26.8	25.0	22.2	16.5	10.0	3.2	(27)
(28)			MCDWB	5.0	5.0	13.8	19.6	24.8	28.4	30.0	29.8	27.4	22.2	14.0	5.1	(28)
(29)		2%	WB	1.1	1.5	7.0	12.6	16.9	22.2	24.6	23.0	19.7	13.9	7.1	1.3	(29)
(30)			MCDWB	2.4	2.5	10.3	18.4	22.2	25.4	28.2	27.9	23.9	18.6	10.0	2.7	(30)
(31)		5%	WB	-1.0	0.2	4.2	10.7	15.4	20.2	23.1	21.6	18.2	11.8	5.0	-0.5	(31)
(32)			MCDWB	-0.2	0.9	6.4	16.3	20.3	24.1	26.6	26.3	22.8	15.9	7.2	0.3	(32)
(33)		10%	WB	-3.7	-2.1	2.2	9.0	14.0	18.8	21.6	20.3	16.6	9.9	3.0	-2.3	(33)
(34)			MCDWB	-3.2	-1.4	3.3	14.1	19.4	23.1	25.4	24.7	20.8	13.5	4.7	-1.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.4	9.8	9.3	11.2	12.4	11.1	11.9	13.1	12.4	10.1	8.6	8.5	(35)
(36)			MCDBR	11.2	10.1	11.3	17.4	18.0	14.6	14.1	16.6	17.2	16.0	13.2	10.1	(36)
(37)			MCWBR	10.2	8.8	8.3	9.8	9.0	7.6	8.4	8.6	9.1	9.7	9.5	9.0	(37)
(38)		5% WB	MCDBR	11.3	9.7	11.3	15.8	13.9	12.3	13.1	14.2	14.7	15.1	12.6	10.1	(38)
(39)			MCWBR	10.3	8.5	8.5	9.7	8.1	7.7	8.7	8.4	8.8	9.5	9.3	8.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.270	0.292	0.302	0.358	0.374	0.380	0.388	0.383	0.348	0.304	0.271	0.251	(40)	
(41)		taud	2.257	2.214	2.359	2.329	2.344	2.355	2.341	2.347	2.436	2.536	2.480	2.375	(41)	
(42)		Ebn at Noon	778	848	911	888	884	879	865	852	851	842	795	767	(42)	
(43)		Edh at Noon	71	96	100	115	119	119	118	112	93	70	58	55	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.45	2.66	3.92	4.99	5.50	5.92	6.28	5.15	3.75	2.27	1.28	1.03	(44)	
(45)		RadStd	0.14	0.20	0.33	0.37	0.44	0.42	0.26	0.34	0.27	0.30	0.17	0.12	(45)	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.91	-1.38	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)		N/A	N/A	N/A	N/A	N/A	-0.72	N/A	N/A	N/A	N/A

Nomenclature: See separate page